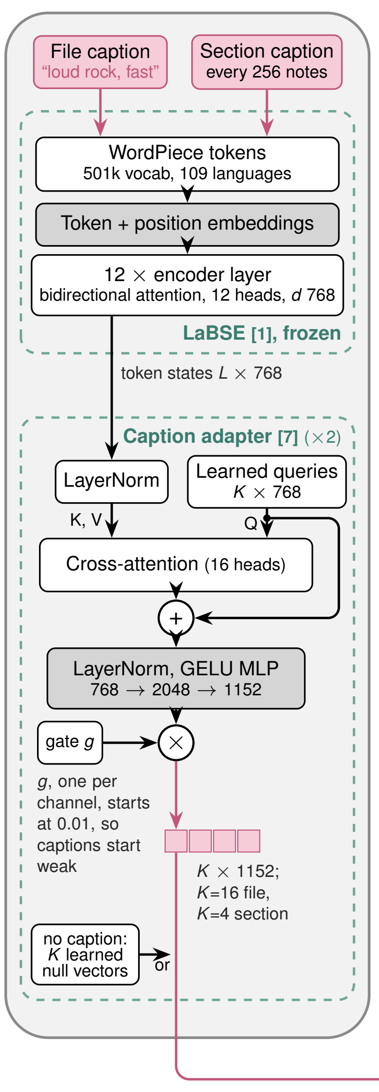
Captioned music model (427M)

427M trainable parameters: trunk 396M, cascade heads 16M, two caption adapters 13M, embeddings 2M. The LaBSE text encoder (471M) is frozen and not counted.

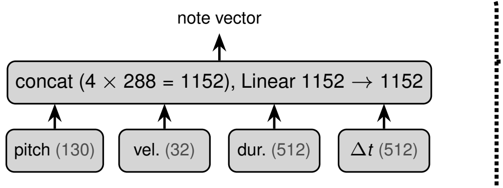
Status: the music model without caption adapters (464M, full attention) is trained. The caption stage of this layout (F1, with GQA) is planned; caption results here come from 107M pilots. F1 will also add a length-control input (roadmap 11), not drawn here yet; infilling (roadmap 12) comes in a later fine-tune.

How to read: numbers give the reading order; pink = captions; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part.

1 Caption path



2 Input embedding

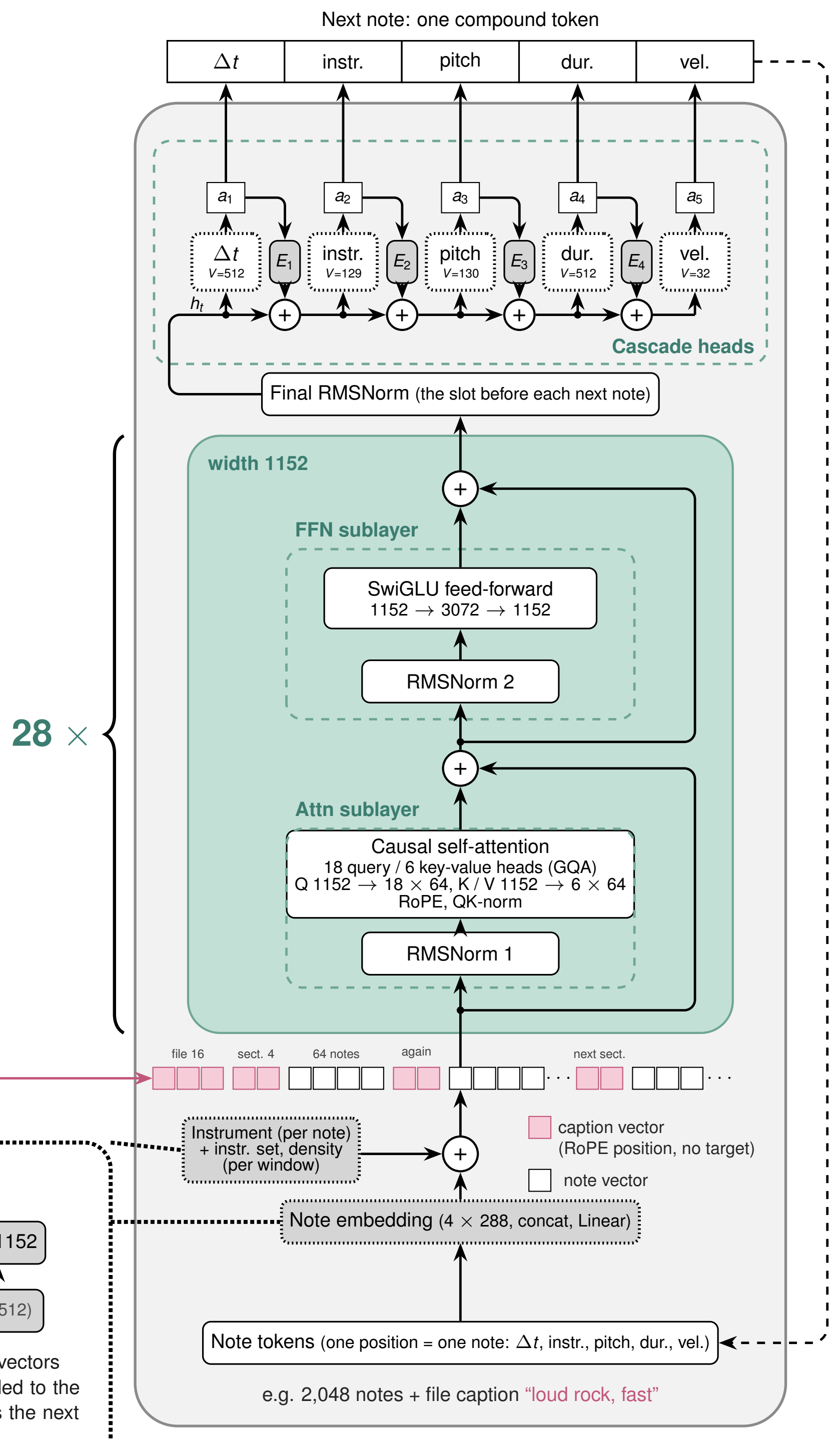


+ instrument of this note: a table of 129 vectors (128 General MIDI programs + drums), added to the note vector. The output head *instr.* predicts the next one.

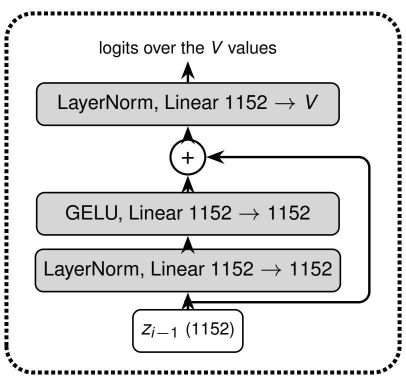
+ window conditions, the same vector at every note position: the **instrument set** of the window (129-dim multi-hot → Linear) and its **density** (notes per second in 15 levels + "none" → a table of 16 vectors). Both start at zero and are dropped 15% of the time in training, so "none" stays usable. In generation they come from the user or are measured on the prompt.

Time is absolute: Δt and durations in 20 ms steps, clipped at 10.22 s. There are no bars, beats or tempo tokens; the metre has to be read from the timing. 2,048 notes are about 1–4 minutes of music.

3 Music model



4 One cascade head (the 5 dotted boxes, own weights)



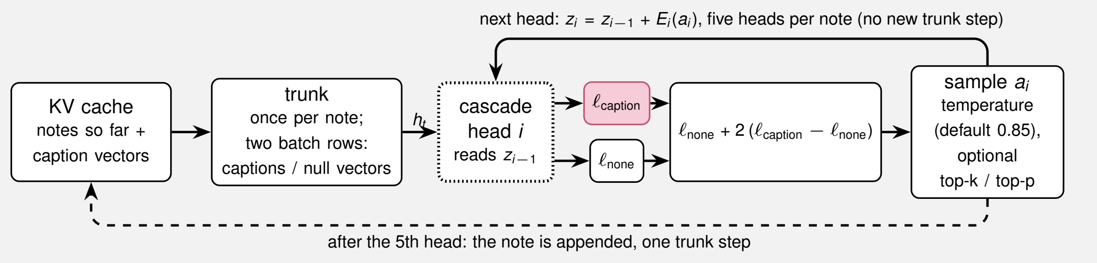
Why a cascade. One position predicts all five values of the next note. Head i reads the running state

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

the trunk state h_t plus the embeddings E of the values already chosen. So the note's probability follows the chain rule, $p(\Delta t)\,p(\text{instr.} \mid \Delta t)\,p(\text{pitch} \mid \Delta t, \text{instr.}) \cdots$ (compound tokens as in [6]).

a_i is the true value in training (teacher forcing) and the sampled value in generation. Loss: the sum of the five cross-entropies. Without the chain: +0.34 nats / note at 42M (section 7).

5 Generation



Classifier-free guidance [9] with scale 2: every step runs the batch twice as wide, once with the captions and once with null vectors.

Caption vectors enter the cache as in training: the file caption first, then a section's 4 vectors at its first note and again every 64 notes. Past 2,048 notes the window slides by a quarter.

6 Training recipe for section captions

Without these measures the model follows the first section's caption and then ignores later ones.

1. Change-aware captions. Each section caption names what changed since the section before ("drums enter, louder"). Half of the training windows contain a section change.

2. Spliced windows (50% of windows).

file A notes | S_B | file B section

File A up to a section boundary, then a section of file B (dashed) with B's caption and no file caption: the caption contradicts the context and has to win.

3. Caption repeated every 64 notes, in training and in generation.

64 notes | 64 notes | 64 notes

The same 4 section vectors are inserted again, so the caption is never far back in the context.

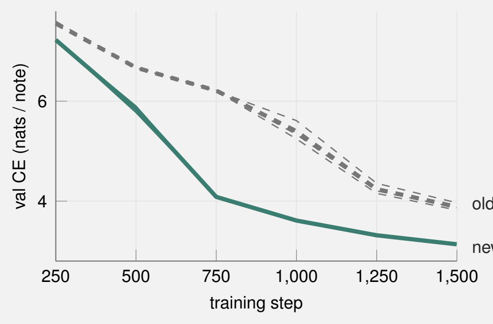
4. Section words from section statistics. Loudness, pace and register words use thresholds over sections, not whole files ("loud" = the loudest 20% of sections), and pace is split into rhythm (onsets per second) and texture (notes per onset), so a section caption does not push the music to extremes.

5. Caption dropout → guidance. File captions are replaced by null vectors 15% of the time, section captions 20%, so generation can use classifier-free guidance [9] with scale 2.

Evidence so far (107M pilot, one seed). Switch test: 6 prompt pairs A/B (e.g. soft / loud), 16 samples each, the section caption changes at note 1,024, guidance 2. The matching attribute (velocity, notes per second, register, drum or string share) is measured per 256 notes; the score is its change after the switch divided by its offset before it (0 = keeps the first style, 1 = a full flip; single prompts reach 1.5). The start model v2_A0: 0.01. 1,500 more steps without measures 2–4 (control): **0.06**, held-out NLL 2.772; splice alone 0.59, repeat alone 0.09. The recipe above (2,700 steps): **0.98**, NLL 2.757. Tested and not used: a counterfactual loss [8] (the true caption must beat the previous one by 0.05 nats on the changed section's first 32 notes): alone 0.13, with splice and repeat 0.85 (0.94 with the section words above), at 1.8× the step time (a second forward). Open cost: holding a caption still overshoots: asked for strings, 34% of samples run away (6.8 notes per onset; 24% and 2.0 with the counterfactual loss). No error bars yet.

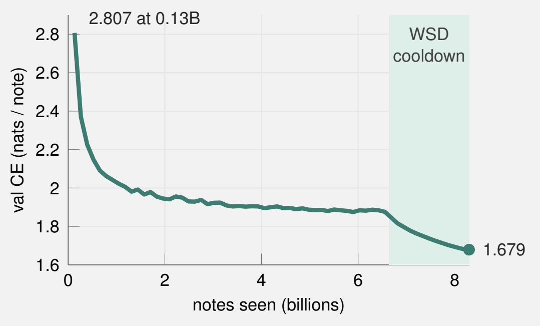
7 Why these choices: training curves (teal = what the model uses)

Trunk block (42M, 98M notes, 3 seeds each)



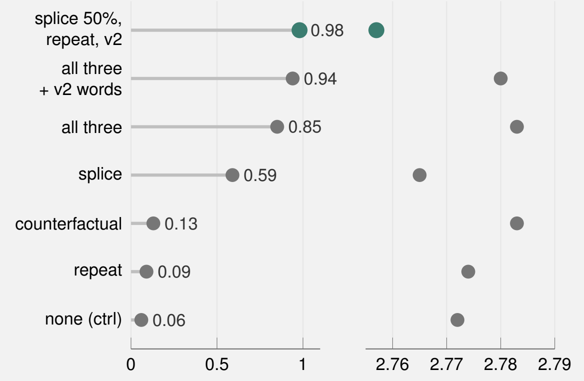
Clean GigaMIDI validation. **New**: RMSNorm, SwiGLU, QK-norm; **old** (dashed): LayerNorm, GELU; both RoPE and Muon. Thin lines = seeds, thick = mean; −0.75 at the end.

Base model pretraining (464M, 8.3B notes)



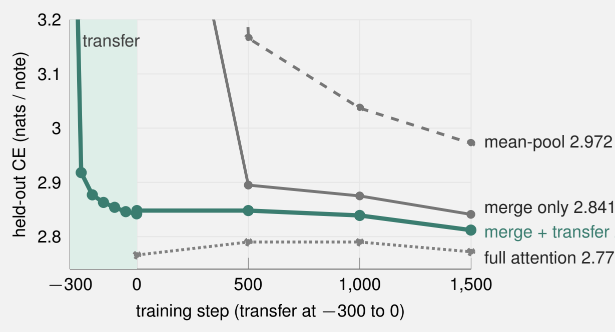
Clean GigaMIDI validation every 2,000 steps (65,536 notes per step); constant LR, then the $1-\sqrt{\cdot}$ cooldown in the last 20%. One run on one RTX 5090, ≈41 h.

Section-caption measures (107M pilot, 1,500–2,700 steps)



Each row continues the same model (v2_A0) for 1,500 steps with one measure or several; the last row runs 2,700 steps, the wall time of "all three" (no second forward). v2 words: measure 4 of section 6. Switch score as in section 6; one seed, no error bars. **Teal** = the recipe in use.

GQA conversion (107M pilot, 12 → 4 key / value heads)



v2_A0 converted, then 1,500 steps on the section recipe's data, LR 10^{-4}, no section measures; the dotted line is the same 1,500 steps without the conversion. Off the scale at the start: mean-pool 10.15, merge 3.86–3.88. One seed.

Why GQA [21], and how a trained model is converted

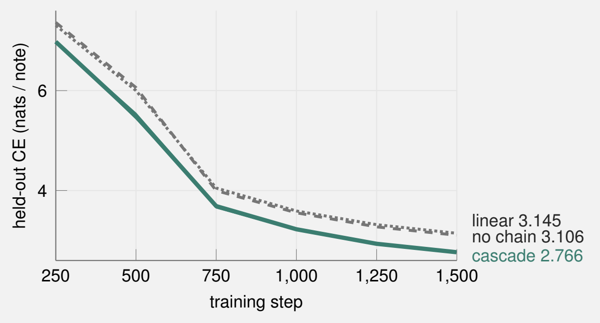
What it buys. Each group of 3 query heads shares one key / value head: the K / V projections and the KV cache shrink 3×, so the 450M has 49.5M fewer parameters (464M → 415M without the caption adapters). Generation reads less per note: the 107M decodes 1.03× faster at 1 row and 2.28× at 64 rows (laptop GPU, bf16); on the Raspberry Pi (int4 weights, int8 cache) the estimate from bytes read per note is ≈1.3× at 1 row and ≈2.1× for 4 candidates with guidance.

Merge. Mean-pooling a group's K / V weights starts at CE 10.15. Instead, each group's value heads are merged by a PCA of their outputs weighted by the output projection, and the key heads per RoPE rotation pair by a rank-1 complex merge weighted by the query energy, so RoPE stays exact. Measured on 16 batches of the model's own inputs.

Transfer. 300 steps in which only the attention layers train, each to reproduce its own full-attention output (relative MSE) on the original layer inputs, LR 10^{-3}; the rest of the model is frozen.

Result. After 1,500 steps: mean-pool +0.200 vs full attention, merge only +0.069, merge + transfer +0.040 and still closing. F1 converts the 450M the same way (18 → 6 heads) and trains ≈30× longer.

Output heads (42M, 98M notes)



Held-out GigaMIDI. **Cascade**: each head reads the values already chosen, 3 seeds (thin) and mean. **No chain** (dashed): the same residual heads, each reading h_t alone; **linear** (dotted): one linear layer per attribute; one seed each. The chain is worth −0.34 at the end (pitch +0.18, velocity +0.11 without it), the residual head form only −0.04.

Per head at step 1,500 (seed 1337, nats / note)

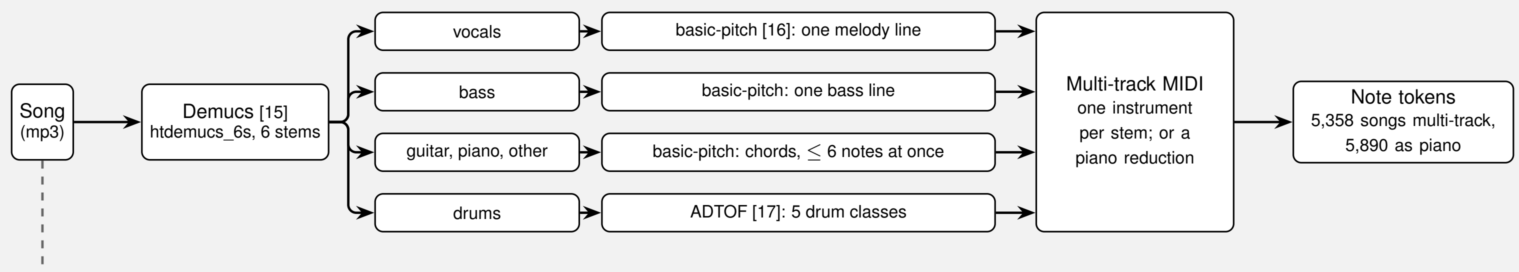
heads	Δt	pitch	dur.	vel.	sum	instr.
cascade	0.285	0.779	0.966	0.737	**2.766**	0.315
no chain	0.281	0.956	1.025	0.844	3.106	0.346
linear	0.286	0.970	1.037	0.853	3.145	0.353

Δt is the first head, so it reads nothing extra and is the same in all three. Pitch gains most from the chain (−0.18): knowing when the note starts and on which instrument narrows its pitch. The sum is the held-out CE above; the instrument head is reported apart. An earlier 6M piano A/B (4 heads, chained plain MLP heads) had found only −0.03 for the chain: there its pitch head had lost the direct path from h_t.

8 Training stages

Setting	Base pretraining (done)	Caption training, F1 run (planned)	Fine-tune on Ukrainian songs
Start	from scratch	the base with its attention made GQA (18 → 6 key / value heads: each group's K / V merged from the data, then 300 steps of per-layer attention transfer) + the two new caption adapters; also planned, not drawn: a length-control input (roadmap 11)	the base (done, no captions); the F1 model with captions (F2, planned)
Data	GigaMIDI [12] 0.4, Aria-MIDI [13] 0.3, Discover MIDI [19] 0.3; 8.3B notes (1.8 epochs on average; Aria ≈4.5)	captioned MIDI in 3 tiers: MidiCaps [14] / XMIDI [20] / GigaMIDI metadata, Aria-MIDI metadata, the rest with feature captions; 2.5–4B notes	5,358 songs transcribed from audio (section 9; 4,811 train, 547 val) 0.75 + GigaMIDI replay 0.25 against forgetting
Captions	none: only the instrument-set and density conditions	templates filled from metadata (genre, mood, artist, composer) and from features computed from the notes (tempo, loudness, register, instruments), 30% in Ukrainian; MidiCaps' LLM-written captions; section captions from each section's own notes	none in the done run; F2 adds them
Window	2,048 notes (≈1–4 min)	2,048 notes + 16 file vectors + 4 per caption arrival (section start and every 64 notes): 2,212 positions (37 caption slots per window, fixed shape; ≈2,192 used)	2,048 notes
Batch	65,536 notes per step (32 windows)	same	24,576 notes per step (12 windows)
Optimizer, LR	Muon [10] for weight matrices, AdamW for the rest; one LR 10^{-3} (Muon updates scaled to AdamW's size)	8-bit AdamW, LR 10^{-4}, adapters 10× (as in the 107M pilots)	8-bit AdamW, LR $2 \cdot 10^{-4}$
Schedule	WSD [11]: warmup, stable (constant LR), $1-\sqrt{\cdot}$ decay over the last 20%	same; planned: captioned data weighted up in the decay (roadmap 13)	same
Steps	126,582 (≈41 h on one RTX 5090)	≈38–61k	1,500
Precision	bf16 with fp8 matmuls, torch.compile	same	same

9 Audio → MIDI: how songs become training notes (fine-tune data)



Every step adds errors: stem bleed, missed or extra notes, octave and timing errors, drum classes mixed up, loudness guessed from activations; stems quieter than −30 dB vs the mix are dropped. The model learns these errors as if they were music.

Planned: a pretrained audio tokenizer (neural codec such as EnCodec [18]) turns the recording itself into tokens, so songs are learned without the transcription noise.

10 Glossary

Δt	time since the previous note started (20 ms steps, 512 values)	**RoPE [2]**	rotary position embedding: positions as rotations of queries and keys
instr.	instrument: General MIDI program 0–127 or drums (129 values)	**RMSNorm [3]**	layer normalisation by the root mean square only
pitch	MIDI pitch 0–127 + start / end of piece (130 values)	**SwiGLU [4]**	feed-forward layer with a SiLU-gated linear unit
dur.	note length (20 ms steps, 512 values)	**QK-norm [5]**	queries and keys normalised before the dot product
vel.	velocity: how hard the note is played (32 levels)	**LaBSE [1]**	multilingual sentence encoder (109 languages)
compound token	one sequence position = one whole note	**NLL**	negative log-likelihood of the true notes
caption vector	a caption-adapter output placed in the sequence; no target of its own, but the last vector of a section caption predicts the next note	**nats**	units of NLL with the natural log
null vectors	learned stand-ins for "no caption"	**BCE**	binary cross-entropy (two-class loss)
teacher forcing	training feeds the true earlier values, not samples	**CFG [9]**	classifier-free guidance: push logits away from the caption-free prediction
KV cache	keys and values of earlier positions kept between steps (GQA: 6 heads per layer, a third of 18)	**WSD [11]**	warmup, stable, decay: LR schedule with a short final cooldown
multi-hot	a 0/1 vector with a 1 for every instrument present	**Muon [10]**	optimizer that orthogonalises the update of each weight matrix
stem	one separated part of a recording (vocals, bass, drums, ...)	**Demucs [15]**	neural source separation: a song into 6 stems
		GQA [21]	grouped-query attention: each group of 3 query heads shares one key / value head

11 References (links)

[1] LaBSE: Language-agnostic BERT Sentence Embedding (2020), `arXiv:2007.01852`
[2] RoFormer: rotary position embedding (2021), `arXiv:2104.09864`
[3] Root Mean Square Layer Normalization (2019), `arXiv:1910.07467`
[4] GLU Variants Improve Transformer (2020), `arXiv:2002.05202`
[5] Query-Key Normalization for Transformers (2020), `arXiv:2010.04245`
[6] Compound Word Transformer (2021), `arXiv:2101.02402`
[7] Flamingo: learned-query resampler, gated adapters (2022), `arXiv:2204.14198`
[8] MuseBarControl: counterfactual loss (2024), `arXiv:2407.04331`
[9] Classifier-Free Diffusion Guidance (2022), `arXiv:2207.12598`
[10] Muon is Scalable for LLM Training (2025), `arXiv:2502.16982`
[11] Scaling laws beyond fixed training durations: WSD (2024), `arXiv:2405.18392`
[12] The GigaMIDI Dataset (2025), `arXiv:2502.17726`
[13] Aria-MIDI: a dataset of piano MIDI files (2025), `arXiv:2504.15071`
[14] MidiCaps: MIDI with text captions (2024), `arXiv:2406.02255`
[15] Demucs: Hybrid Transformers for Music Source Separation (2022), `arXiv:2211.08553`
[16] basic-pitch: lightweight polyphonic note transcription (2022), `arXiv:2203.09893`
[17] ADTOF: automatic drum transcription dataset and model (2021), `arXiv:2111.11737`
[18] EnCodec: High Fidelity Neural Audio Compression (2022), `arXiv:2210.13438`
[19] Discover MIDI Dataset, `Hugging Face dataset`
[20] XMusic: the XMIDI dataset (2025), `arXiv:2501.08809`
[21] GQA: multi-query models from multi-head checkpoints (2023), `arXiv:2305.13245`